

AIR (Advice & Intelligence Relationship) System Architecture Design – V2.0

Version: 2.0 – BIAN-Aligned Architecture

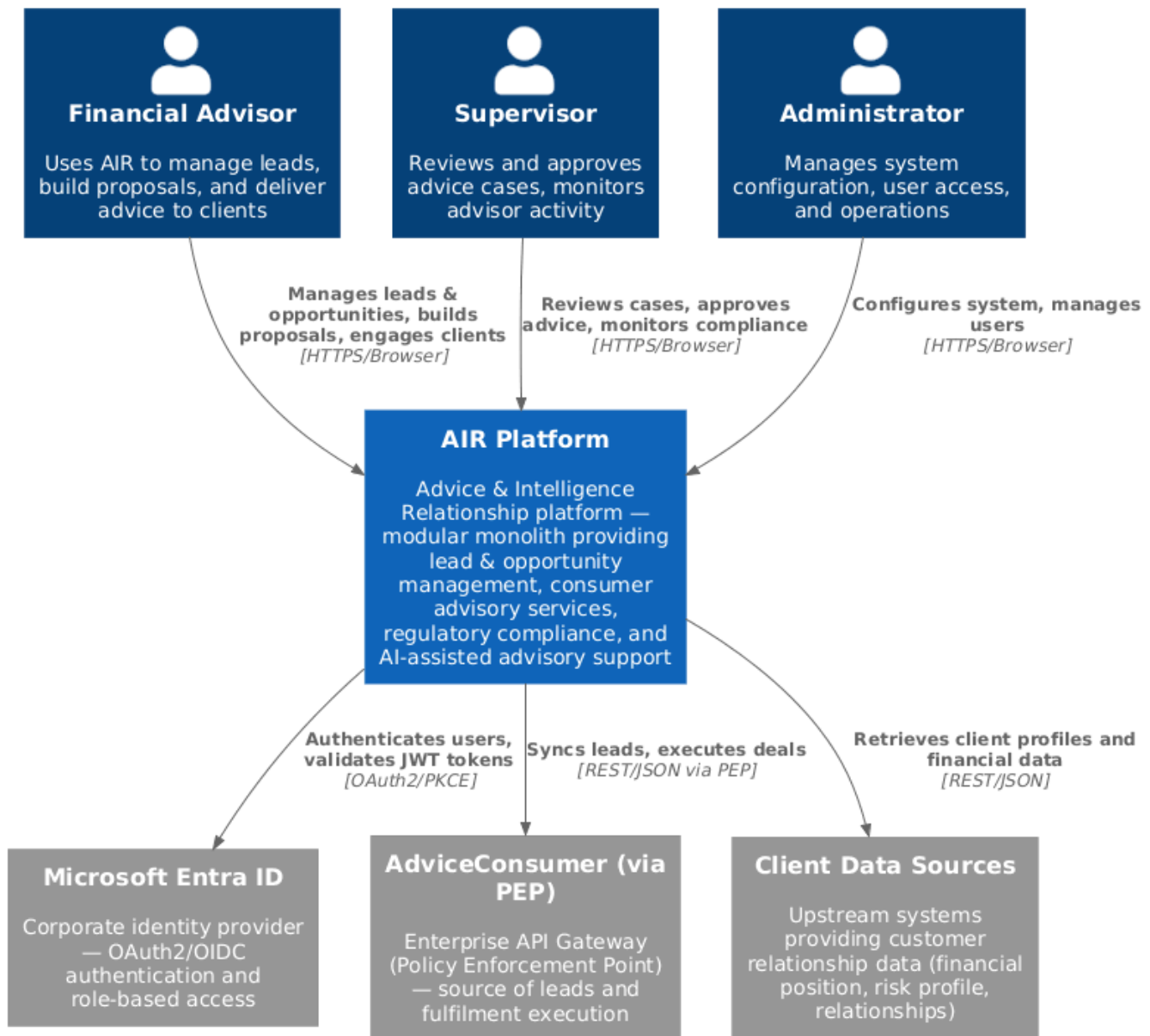
Based on: System Architecture Design V1.0 + AIR BIAN-Aligned DDD v3

Date: May 2026

Level 1: System Context Diagram

Shows the AIR system in its environment, with external actors and systems.

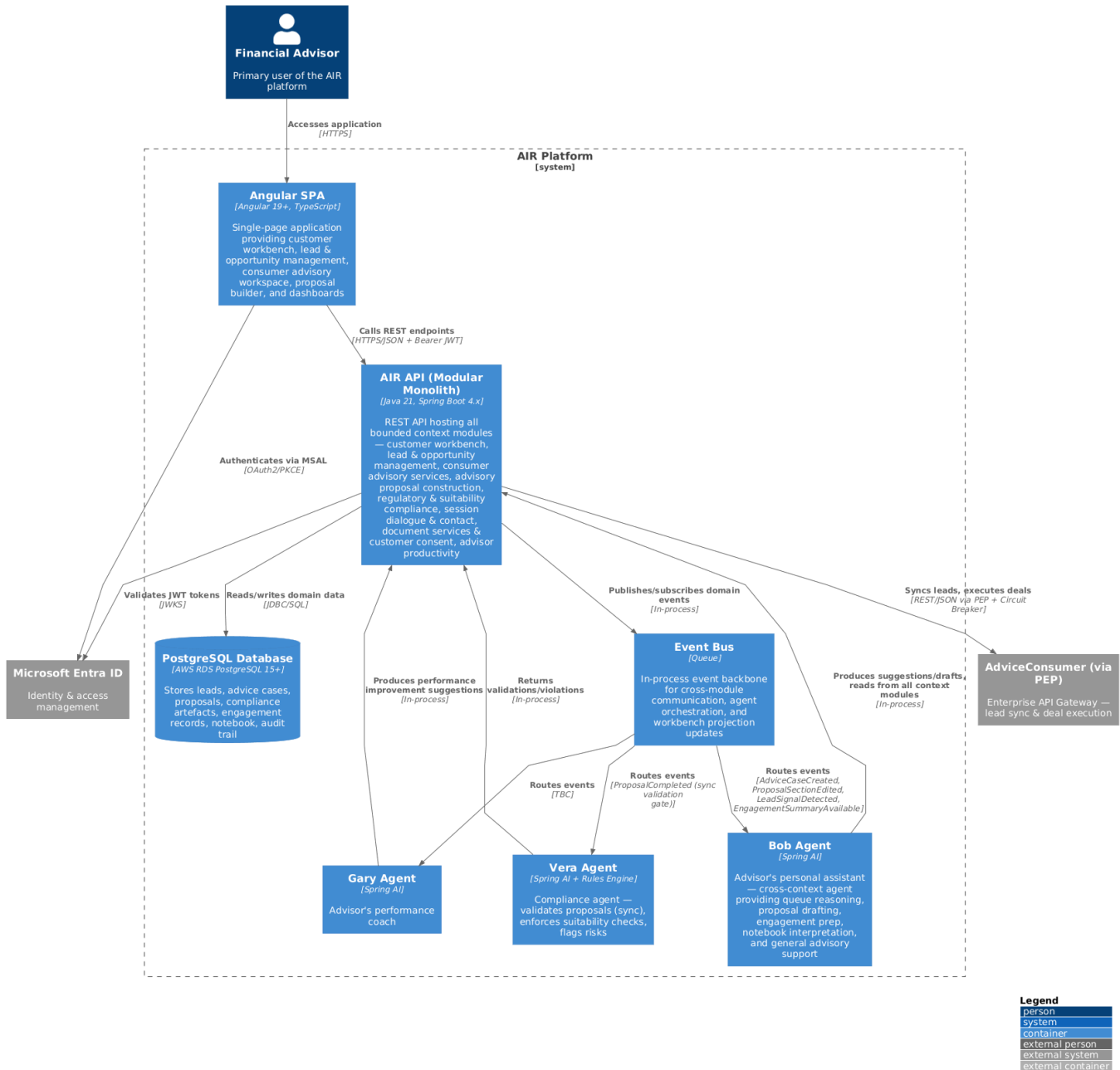
AIR System Context Diagram



Level 2: Container Diagram

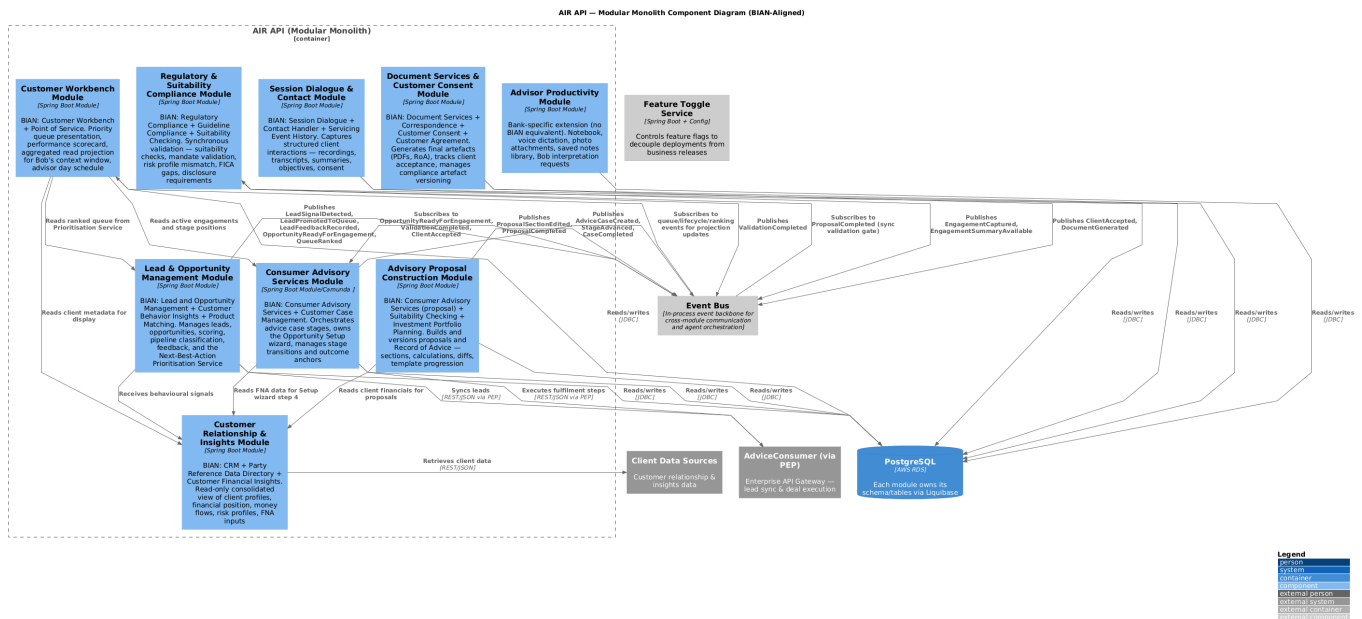
Shows the major deployable units and their interactions.

AIR Container Diagram



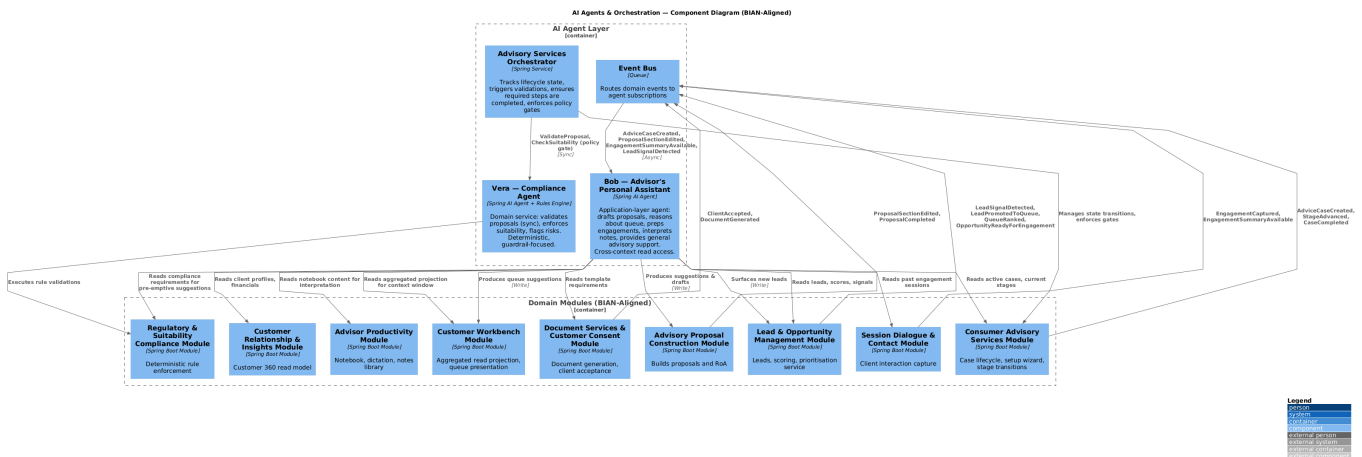
Level 3: Component Diagram – AIR API (Modular Monolith)

Shows the internal module structure of the monolith, aligned to the BIAN v14 bounded contexts.



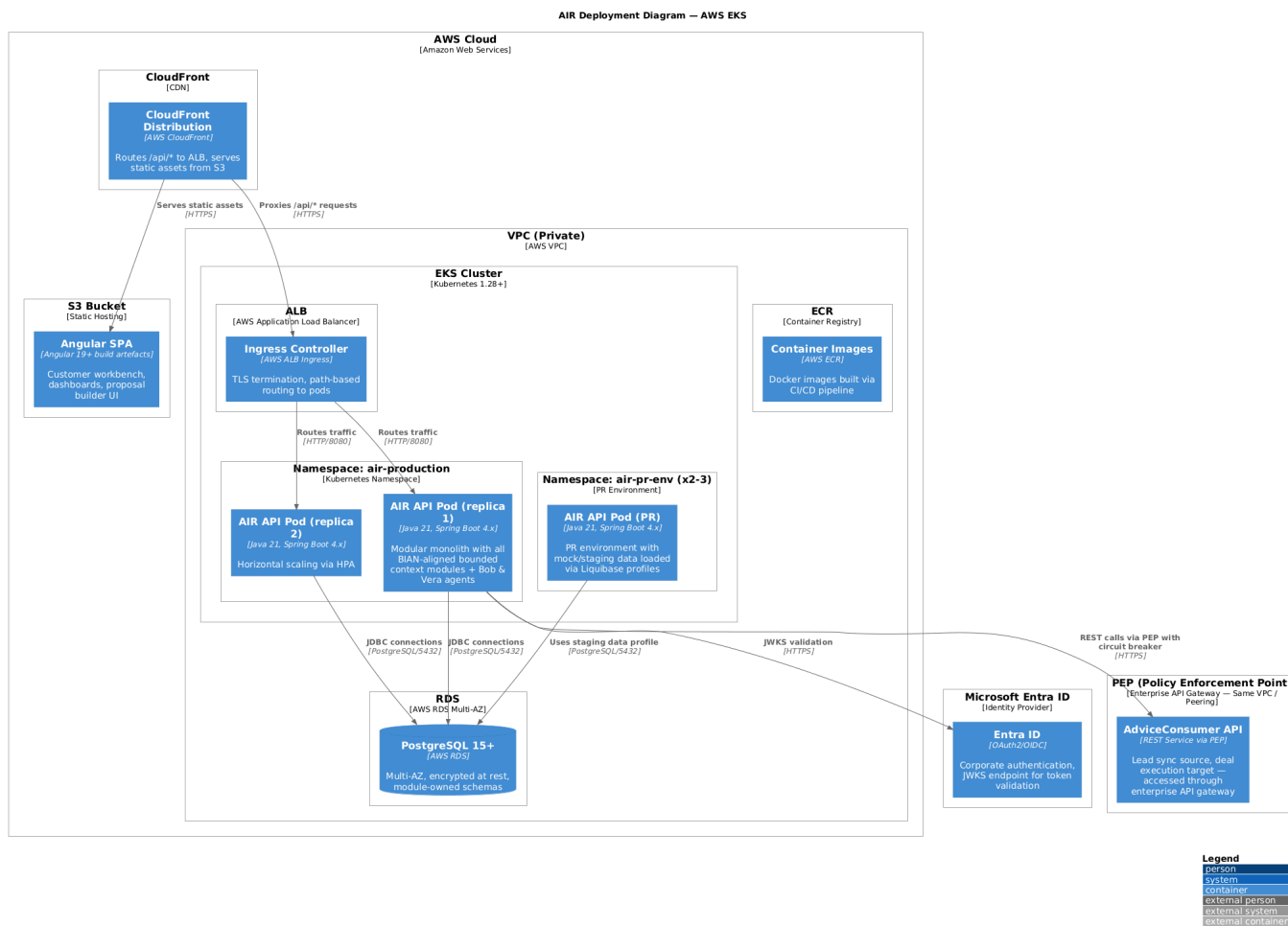
Level 3: Component Diagram – AI Agents & Orchestration

Focused view of how Bob and Vera interact with the BIAN-aligned domain modules. Gary is deferred.



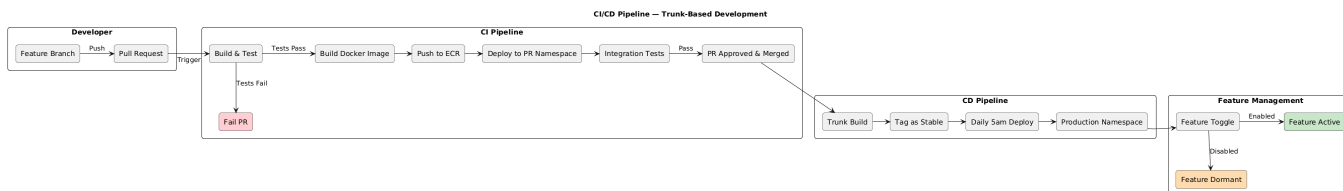
Level 4: Deployment Diagram

Shows the physical deployment topology on AWS. Unchanged from V1.0 – deployment is infrastructure-level and independent of domain naming.



CI/CD Pipeline Flow

Illustrates the trunk-based development and deployment pipeline. Unchanged from V1.0.



Event Flow Diagram

Shows the event-driven orchestration pattern aligned to the BIAN v14 bounded contexts.

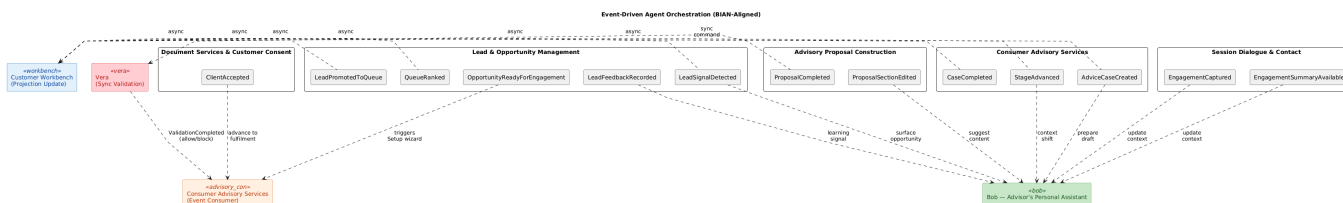


Diagram Summary

Level	Diagram	Purpose
C4 Level 1	System Context	Shows AIR in its ecosystem with users and external systems
C4 Level 2	Container	Shows deployable units – SPA, API monolith, DB, Bob & Vera agents, event bus
C4 Level 3	Component (API)	Shows internal BIAN-aligned bounded context modules within the monolith (9 active contexts)
C4 Level 3	Component (Agents)	Shows Bob's cross-context read/write access and Vera's sync validation role
C4 Level 4	Deployment	Shows AWS infrastructure topology – EKS, RDS, CloudFront, S3
Supplementary	CI/CD Pipeline	Shows trunk-based development and daily deployment flow
Supplementary	Event Flow	Shows event-driven orchestration with BIAN-aligned event catalogue

Key Architecture Characteristics

- **Modular Monolith:** All bounded contexts deployed as a single unit for speed, with explicit module boundaries for future decomposition
- **9 Active Bounded Contexts (BIAN-Aligned):** Customer Workbench, Lead & Opportunity Management, Customer Relationship & Insights, Consumer Advisory Services, Advisory Proposal Construction, Regulatory & Suitability Compliance, Session Dialogue & Contact, Document Services & Customer Consent, Advisor Productivity
- **BIAN v14 Naming Convention:** All contexts use canonical BIAN service domain names where strong alignment exists; bank-specific extensions are clearly documented
- **Bob as Application-Layer Agent:** Cross-context personal assistant with read access to all modules and write access to Advisory Proposal Construction, Customer Workbench, and Lead & Opportunity Management
- **Vera as Domain Service:** Synchronous compliance validation within the Regulatory & Suitability Compliance module; async monitoring deferred
- **Gary Deferred:** Professional coach agent deferred to future phase (Servicing Activity Analysis context)
- **Prioritisation Service in Lead & Opportunity Management:** The Next-Best-Action ranking algorithm lives as a domain service within the Lead & Opportunity module, publishing ranked results to the Customer Workbench projection

- **Opportunity Setup Wizard in Consumer Advisory Services:** Triggered by `OpportunityReadyForEngagement` event from Lead & Opportunity Management, owned by Consumer Advisory Services
- **Event-Driven Backbone:** 13 domain events orchestrate cross-module communication and agent behaviour
- **Policy Gates:** Critical state transitions (e.g. `CompleteProposal`) go through synchronous Vera validation
- **AI Cannot Change State:** Agents suggest, flag, and advise – humans approve and transition
- **Feature Toggles:** Decouple code deployment from business release
- **PR Environments:** Ephemeral namespaces with mock data for rapid feedback
- **Module-Owned Data:** Each module manages its own schema via Liquibase

Alignment to BIAN-Aligned DDD v3

Bounded Context (BIAN-Aligned)	Module	BIAN Service Domain(s)	Classification	Notes
Customer Workbench	<code>workbenchModule</code>	Customer Workbench, Point of Service	Core	Owns queue, presentation, scorecard, aggregate, read projection
Lead & Opportunity Management	<code>leadOpportunityModule</code>	Lead and Opportunity Management, Customer Behavior Insights, Product Matching	Core	Owns lead scoring, pipeline, prioritisation, service, feedback
Customer Relationship & Insights	<code>customerInsightsModule</code>	CRM, Party Reference Data Directory, Customer Financial Insights	Core (read model)	Read-only projection, upstream client sync
Consumer Advisory Services	<code>advisoryServicesModule</code>	Consumer Advisory Services, Customer Case Management	Core	Owns state transition, setup wizard, outcome & feedback

Bounded Context (BIAN-Aligned)	Module	BIAN Service Domain(s)	Classification	Notes
Advisory Proposal Construction	proposalModule	Consumer Advisory Services (proposal), Suitability Checking, Investment Portfolio Planning	Core	Owns proposals, RoA, versioning, diffs
Regulatory & Suitability Compliance	complianceModule	Regulatory Compliance, Guideline Compliance, Suitability Checking	Core	Sync validation only; async monitoring deferred
Session Dialogue & Contact	sessionDialogueModule	Session Dialogue, Contact Handler, Servicing Event History	Supporting	Structure interactions, capture 1 to cases
Document Services & Customer Consent	documentServicesModule	Document Services, Correspondence, Customer Consent, Customer Agreement	Supporting	Document generation, acceptance tracking, compliance artefacts
Advisor Productivity	productivityModule	(No BIAN equivalent – bank-specific extension)	Supporting	Notebook, dictation notes lib

Deferred Contexts (not represented in current architecture)

Context (BIAN-Aligned)	BIAN Service Domain(s)	Reason	Future Home
Party Authentication + Employee Access	Party Authentication, Employee Access	Consumed from bank IAM (Entra ID)	External dependency
Compliance Reporting	Compliance Reporting, Regulatory Reporting	Vera's async mode not yet validated	Future module + event subscriptions

Context (BIAN-Aligned)	BIAN Service Domain(s)	Reason	Future Home
Servicing Activity Analysis (Gary)	Servicing Activity Analysis, Employee Evaluation	Coach agent not in MVP scope	Future module + agent
Deal Execution & Fulfilment	<i>(Absorbed into Consumer Advisory Services)</i>	Terminal lifecycle stage	Extract when complexity warrants

BIAN Naming Convention

All bounded contexts and their corresponding modules use the following naming hierarchy:

1. **Primary BIAN service domain name** – used as the canonical label in API contracts, documentation, and integration specifications
2. **AIR qualifier** (where needed for disambiguation) – appended in parentheses, e.g. "Consumer Advisory Services (Lifecycle)" vs "Consumer Advisory Services (Proposal)"
3. **Bank-specific extension label** – used for contexts with no BIAN equivalent, clearly marked as extensions

Where AIR Extends Beyond BIAN v14

Extension	AIR Module	Rationale
AI Agent orchestration pattern	Bob, Vera, Gary	BIAN models capabilities, not cross-cutting agent constructs
Next-Best-Action prioritisation	Lead & Opportunity Management (domain service)	BIAN handles lead progression but not ranking algorithms
Personal productivity tools	Advisor Productivity	BIAN excludes internal employee productivity tooling
Event-driven backbone	Cross-cutting	BIAN models capabilities, not the event infrastructure connecting them

Glossary

Term	Definition
PEP	Policy Enforcement Point – the enterprise API gateway through which external service calls are routed
AIR	Advice & Intelligence Relationship – the platform being designed
BIAN	Banking Industry Architecture Network – industry-standard reference model for banking service domains
Bob	AI personal assistant agent – advisor's cross-context co-pilot (proposal drafting, queue reasoning, engagement prep, notebook interpretation)
Vera	AI compliance agent – synchronous rule validation and suitability enforcement
Gary	AI professional coach agent – behavioural analysis and coaching insights (deferred)
RoA	Record of Advice – regulatory documentation of advice given
FNA	Financial Needs Analysis – client assessment inputs
HPA	Horizontal Pod Autoscaler – Kubernetes auto-scaling mechanism
NBA	Next-Best-Action – the prioritised recommendation for what an advisor should do next
QTD	Quarter-to-Date – performance measurement period
GAP	Pre-signature opportunity (not yet in pipeline)
Pipeline	Post-signature, pre-fulfilment opportunity (committed revenue)
CRM	Customer Relationship Management – BIAN service domain for managing customer relationships

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